

BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```

Activating project at `~/Desktop/hw2-matt-b`
Installed InvertedIndices          v1.3.1
Installed JpegTurbo_jll            v3.1.3+0
Installed Libmount_jll             v2.41.1+0
Installed InlineStrings            v1.4.5
Installed Accessors                v0.1.42
Installed Roots                    v2.2.10
Installed Preferences              v1.5.0
Installed DataFrames               v1.8.0
Installed Fontconfig_jll           v2.17.1+0
Installed WorkerUtilities          v1.6.1
Installed SentinelArrays           v1.4.8
Installed WeakRefStrings           v1.4.2
Installed DataStructures           v0.19.1
Installed StatsBase                v0.34.6
Installed TableTraits              v1.0.1
Installed Expat_jll                v2.7.1+0
Installed PooledArrays             v1.4.3
Installed DataValueInterfaces      v1.0.0
Installed ConstructionBase         v1.6.0
Installed Ghostscript_jll          v9.55.1+0
Installed PrettyTables             v3.0.8
Installed Libuuid_jll              v2.41.1+0
Installed Crayons                  v4.1.1
Installed CompositionsBase         v0.1.2
Installed InverseFunctions          v0.1.17
Installed CommonSolve              v0.2.4
Installed Latexify                 v0.16.10
Installed Plots                    v1.40.19
Installed OpenSSL_jll              v3.5.2+0
Installed IteratorInterfaceExtensions v1.0.0
Installed Glib_jll                 v2.86.0+0
Installed Tables                   v1.12.1
Installed FilePathsBase            v0.9.24
Installed StringManipulation       v0.4.1
Installed SortingAlgorithms        v1.2.2
Installed CSV                      v0.10.15
Precompiling project...
 621.4 ms   WorkerUtilities
 623.2 ms   DataValueInterfaces
 624.1 ms   CommonSolve
 686.1 ms   IteratorInterfaceExtensions
 625.0 ms   InverseFunctions

```

271.7 ms	CompositionsBase
274.4 ms	InvertedIndices
317.2 ms	ConstructionBase
468.8 ms	InlineStrings
396.6 ms	PooledArrays
806.3 ms	Crayons
487.0 ms	Preferences
936.7 ms	SentinelArrays
599.4 ms	TableTraits
754.8 ms	FilePathsBase
365.9 ms	InverseFunctions → InverseFunctionsDatesExt
1375.5 ms	DataStructures
695.7 ms	InverseFunctions → InverseFunctionsTestExt
347.4 ms	LogExpFunctions → LogExpFunctionsInverseFunctionsExt
534.4 ms	Unitful → InverseFunctionsUnitfulExt
407.8 ms	CompositionsBase → CompositionsBaseInverseFunctionsExt
519.6 ms	ConstructionBase → ConstructionBaseLinearAlgebraExt
527.3 ms	Unitful → ConstructionBaseUnitfulExt
381.6 ms	PrecompileTools
660.1 ms	JLLWrappers
600.5 ms	FilePathsBase → FilePathsBaseMmapExt
885.1 ms	Tables
598.0 ms	SortingAlgorithms
998.5 ms	FilePathsBase → FilePathsBaseTestExt
1312.3 ms	RecipesBase
1347.4 ms	StringManipulation
1948.2 ms	Accessors
372.9 ms	Graphite2_jll
560.8 ms	OpenSSL_jll
328.7 ms	Libmount_jll
351.3 ms	EpollShim_jll
356.4 ms	LLVMOpenMP_jll
392.2 ms	Bzip2_jll
341.3 ms	Xorg_libICE_jll
340.1 ms	Xorg_libXau_jll
426.3 ms	libpng_jll
370.7 ms	libfdk_aac_jll
393.7 ms	LAME_jll
369.4 ms	LERC_jll
454.3 ms	fzf_jll
707.8 ms	JpegTurbo_jll
720.5 ms	XZ_jll
3455.6 ms	ColorSchemes

599.0 ms	Ogg_jll
427.5 ms	mtdev_jll
375.4 ms	Xorg_libXdmcp_jll
390.4 ms	x265_jll
387.7 ms	x264_jll
448.4 ms	libaom_jll
477.0 ms	Zstd_jll
459.4 ms	Expat_jll
448.2 ms	LZO_jll
458.4 ms	Opus_jll
397.0 ms	Xorg_xtrans_jll
390.7 ms	libevdev_jll
435.2 ms	Libiconv_jll
369.3 ms	eudev_jll
366.5 ms	Libuuid_jll
439.5 ms	Libffi_jll
424.5 ms	FriBidi_jll
872.8 ms	Accessors → LinearAlgebraExt
704.4 ms	Accessors → TestExt
591.5 ms	Accessors → UnitfulExt
770.1 ms	Pixman_jll
2542.8 ms	StatsBase
655.6 ms	FreeType2_jll
404.3 ms	Xorg_libSM_jll
2237.2 ms	OpenSSL
912.5 ms	JLFFzf
9458.9 ms	Parsers
607.6 ms	libvorbis_jll
1501.8 ms	Ghostscript_jll
346.0 ms	Xorg_libxcb_jll
344.1 ms	DBus_jll
362.8 ms	libinput_jll
776.0 ms	GettextRuntime_jll
1063.5 ms	Libtiff_jll
355.8 ms	Wayland_jll
1262.2 ms	Fontconfig_jll
2240.5 ms	Roots
1228.9 ms	JSON
548.0 ms	InlineStrings → ParsersExt
371.5 ms	Xorg_xcb_util_jll
408.9 ms	Xorg_libX11_jll
896.2 ms	Glib_jll
706.1 ms	Roots → RootsUnitfulExt

7343.3 ms	PlotUtils
2776.0 ms	Latexify
361.5 ms	Xorg_xcb_util_image_jll
369.5 ms	Xorg_xcb_util_keysyms_jll
675.9 ms	WeakRefStrings
346.5 ms	Xorg_xcb_util_renderutil_jll
338.7 ms	Xorg_libXrender_jll
354.6 ms	Xorg_xcb_util_wm_jll
342.7 ms	Xorg_libXext_jll
359.4 ms	Xorg_libXfixes_jll
354.0 ms	Xorg_libxkbfile_jll
1118.9 ms	UnitfulLatexify
655.2 ms	Latexify → SparseArraysExt
2329.3 ms	PlotThemes
455.4 ms	Xorg_xcb_util_cursor_jll
485.9 ms	Xorg_libXinerama_jll
469.4 ms	Xorg_libXrandr_jll
3638.0 ms	RecipesPipeline
409.2 ms	Libglvnd_jll
469.1 ms	Xorg_libXcursor_jll
428.6 ms	Xorg_libXi_jll
479.3 ms	Xorg_xkbcomp_jll
652.0 ms	Xorg_xkeyboard_config_jll
2448.1 ms	Cairo_jll
412.1 ms	xkbcommon_jll
416.4 ms	Vulkan_Loader_jll
923.1 ms	HarfBuzz_jll
14044.5 ms	HTTP
1052.8 ms	libass_jll
607.9 ms	Pango_jll
375.4 ms	libdecor_jll
408.6 ms	GLFW_jll
3209.5 ms	Qt6Base_jll
2573.9 ms	FFMPEG_jll
1174.5 ms	Qt6ShaderTools_jll
1021.5 ms	FFMPEG
1513.9 ms	GR_jll
1443.7 ms	Qt6Declarative_jll
276.1 ms	Qt6Wayland_jll
11671.1 ms	CSV
25553.7 ms	PrettyTables
2365.0 ms	GR
26480.8 ms	DataFrames

```
1529.4 ms    Latexify → DataFramesExt
35823.1 ms   Plots
2598.6 ms    Plots → UnitfulExt
137 dependencies successfully precompiled in 75 seconds. 82 already precompiled.
```

```
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO₂ concentrations, and ocean CO₂ concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

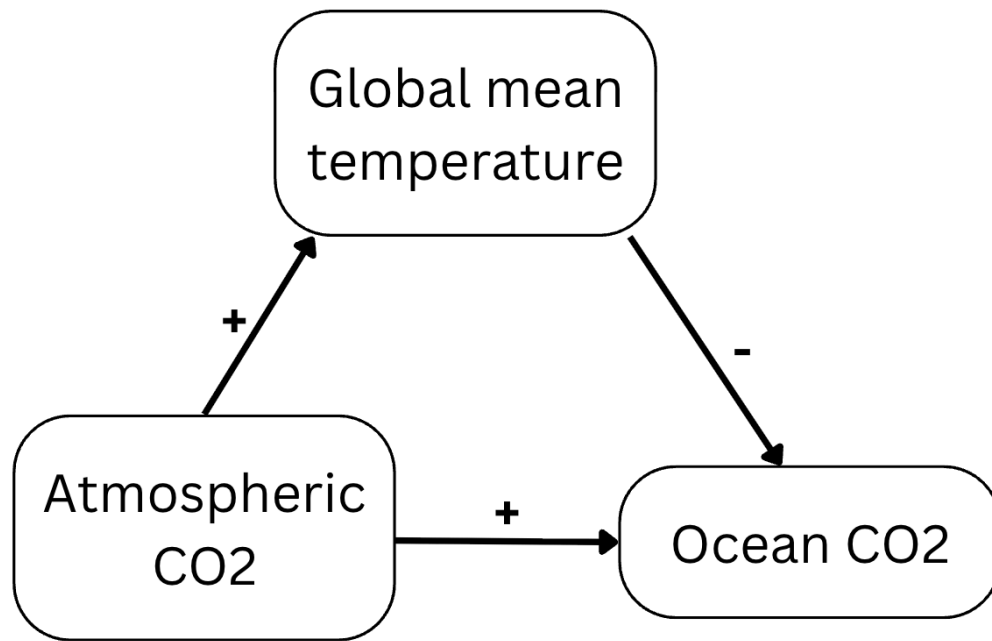


Figure 1: Screenshot 2025-09-21 at 3.24.30 PM.png

As atmospheric CO₂ increases, global mean temperature also increases due to climate change and the increase of the greenhouse effect, which increases that amount of solar radiation that stays within the atmosphere.

As atmospheric CO₂ increases, ocean CO₂ also increases because of Henry's law, which states that the amount of CO₂ in liquid is proportional to its amount in the gas directly above the liquid. (Source: https://en.wikipedia.org/wiki/Henry%27s_law)

As global mean temperature increases, ocean CO₂ decreases because of water solubility. Increased temperature leads to decreased solubility of CO₂, decreasing the proportion of CO₂ that is stored in the ocean. (Source: https://www.engineeringtoolbox.com/gases-solubility-water-d_1148.html)

Overall, this means that there is a negative feedback between temperature and the ocean carbon cycle. Increasing atmospheric CO₂ increases ocean CO₂, while increasing temperature (caused by increasing atmospheric CO₂) decreases ocean CO₂. Henry's law and water solubility push the ocean CO₂ concentration in different directions. If someone were to measure ocean

CO₂ only considering atmospheric CO₂ and disregarding the effects of temperature, ocean CO₂ concentrations would be lower than they would expect.

Tip

Think about Henry's law for CO₂.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

Tip

Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system is in compliance with a regulatory limit of $2.3 \text{ kg}/(1000\text{m}^3)$. You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized “Snowball Earth”. In this problem, we’ll introduce a simple model of the Earth’s energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C \frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1 - \alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}} + \underbrace{a \ln \left(\frac{[CO_2]}{[CO_2]_{PI}} \right)}_{\text{greenhouse effect}}, \quad (1)$$

where:

- T is the Earth’s global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be $51 \text{ J/m}^2/\text{°C}$;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3 \text{ W/m}^2/\text{°C}$ (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2 \text{ W/m}^2$ (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1 \text{ yr}$.

Problem 3.2

Rather than assuming a constant value for the albedo α , we will represent the ice-albedo feedback by letting α depend on T :

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T + 10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoproterozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C ?

References

List any external references consulted, including classmates.